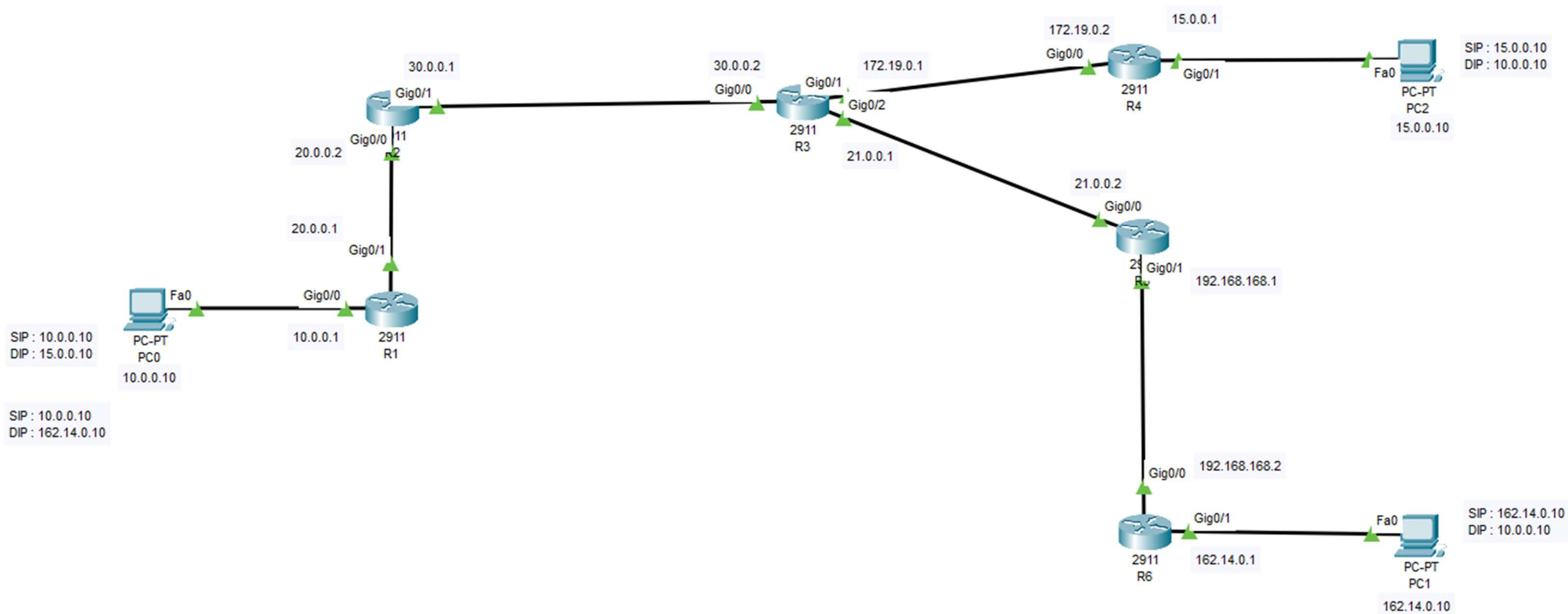
PROJECT 1 – STATIC ROUTE

Network Topology:



Total Network ID in the Topology

1. 10.0.0.0
2. 15.0.0.0
3. 20.0.0.0
4. 21.0.0.0
5. 30.0.0.0
6. 162.0.0.0
7. 172.0.0.0
8. 192.0.0.0

Ensure total 8 network id interface (IP) route is available in each router for the network reachability.

Check route is available for destination IP (10.0.0.1) in each router till you reach 15.0.0.10 and 162.14.0.10 PCs for sending and receiving data.

Let's note all the assigned IP as per network topology and then assign IP interface each router wise. Once interface assigned then automatically directly connected routs will be created in the route table.

Next check the network reachability router by router while sending and receiving from each computer. Then note down missing IP each router wise

Check IP interface in each router

```
R1#show ip int bri
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Configure IP Interface in the Routers as per the topology

```
R1#conf t
R1#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int gig0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int gig0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#
```

Check Interface status

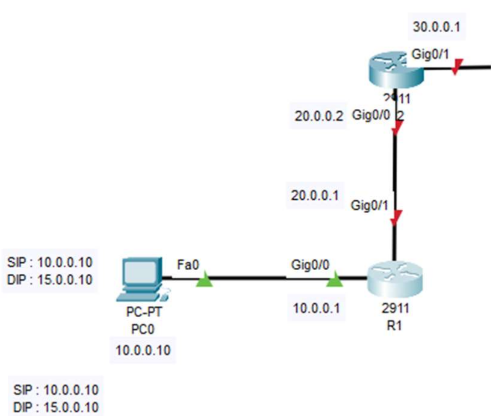
```
R1#show ip int bri
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     20.0.0.1        YES manual up                    down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Gig0/1 protocol is down because there is no route to next router

Check IP Route status

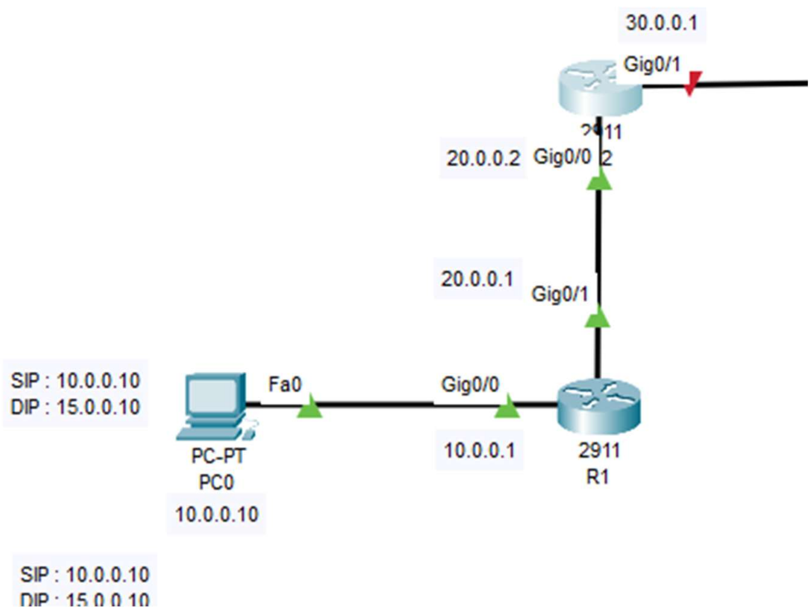
```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0

R1#
```

Gig0/1 network route not showing in the route table because there is no route to (20.0.0.2) in the next router



After interface assigned in Router 2

```
R2#conf t
R2#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int gig0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#exit
R2(config)#
```



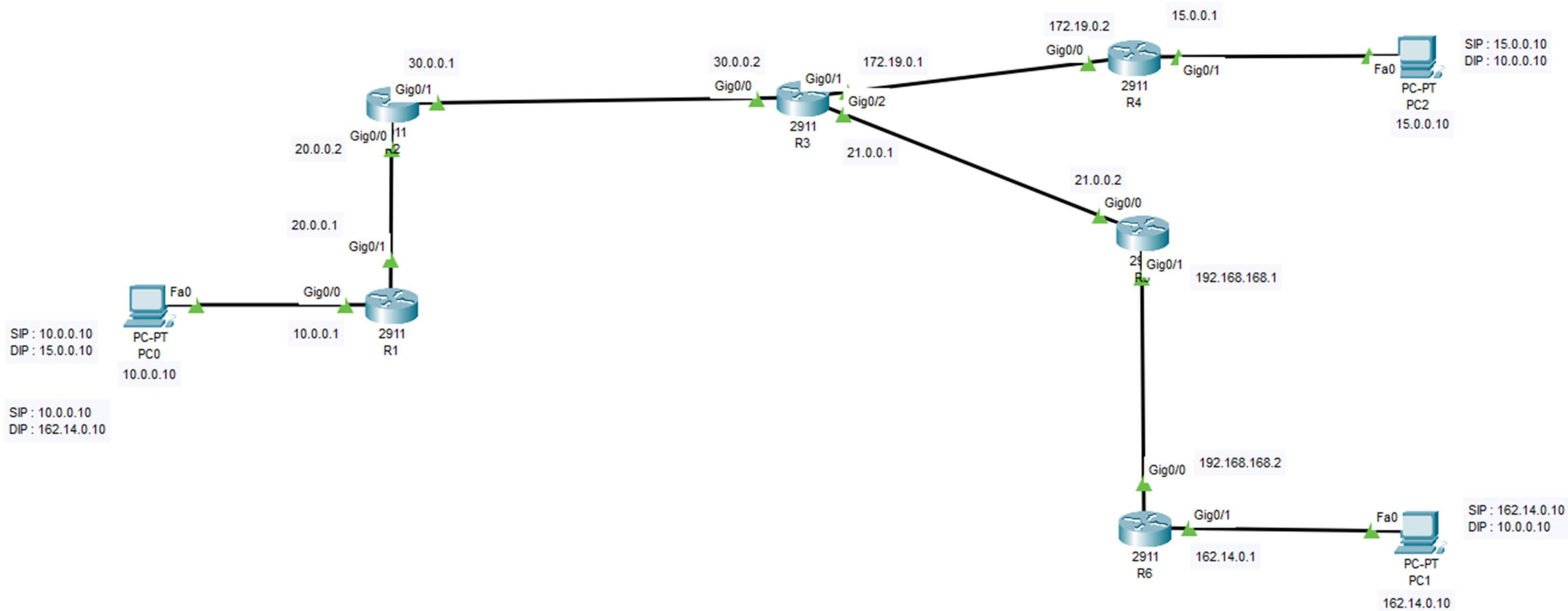
Router 1 IP interface showing now 20 network automatically in the route table

```
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1

R1#
```

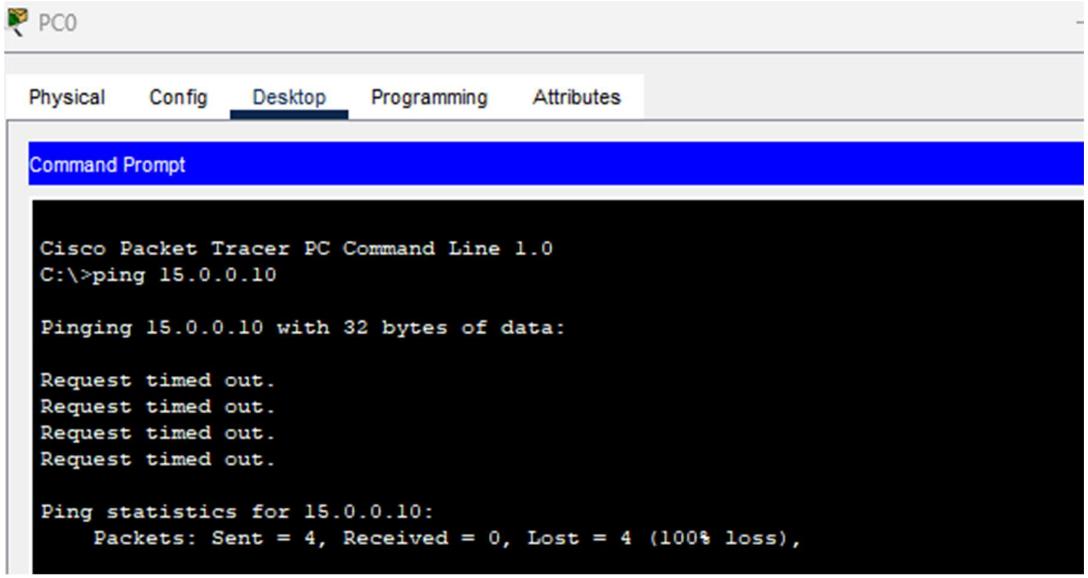
Like wise assigned all the IP interface in all the routers as shown below table.

Configure IP Interface in all the Routers

R1		R2		R3		R4		R5		R6	
IP	Prefix	IP	Prefix	IP	Prefix	IP	Prefix	IP	Prefix	IP	Prefix
10.0.0.1	\8	20.0.0.2	\8	30.0.0.2	\8	172.19.0.2	\16	21.0.0.2	\8	192.168.168.2	\24
20.0.0.1	\8	30.0.0.1	\8	172.19.0.1	\16	15.0.0.1	\8	192.168.168.1	\24	162.14.0.1	\16
				21.0.0.1	\8						

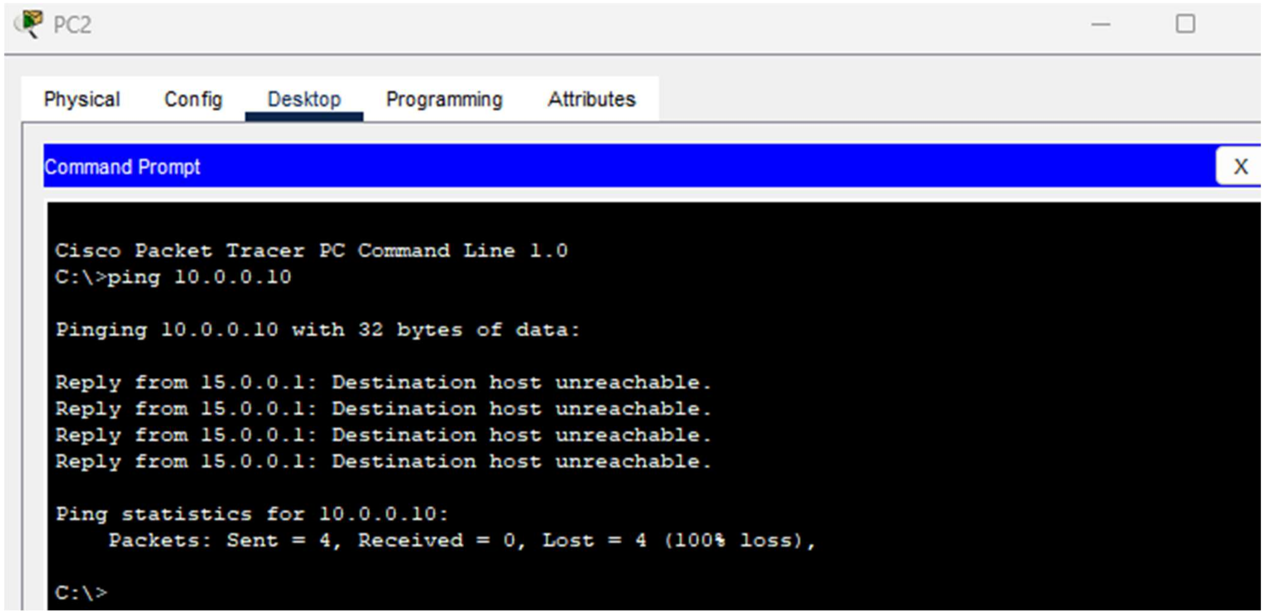


Assign IP add in NIC at PC0 and PC1 and then check ping



Check sends and replay packet side from 10.0.0.10 to 15.0.0.10 and 15.0.0.10 to 10.0.0.10. missing the network id in all the routers as static routs.

```
R1#conf t
R1#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
S    15.0.0.0/8 [1/0] via 20.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
```



Check from 15.0.0.10 destination IP (10.0.0.10) is available in all the routers. We need to assign 10 networks from R4 router to R2.

Check Destination IP Route in each router (15.0.0.10) if not, add static route with network id and netmask and gateway (next hop)

Static Route From 10.0.0.10 to 15.0.0.10

R1			R2			R3			R4		
Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway
15.0.0.0	255.0.0.0	20.0.0.2	15.0.0.0	255.0.0.0	30.0.0.2	15.0.0.0	255.0.0.0	172.19.0.2			

Static Route From 15.0.0.10 to 10.0.0.10

R1			R2			R3			R4		
Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway
						10.0.0.0	255.0.0.0	30.0.0.1	10.0.0.0	255.0.0.0	172.19.0.1

Test ping from 10.0.0.10

```
C:\>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Now successfully getting reply

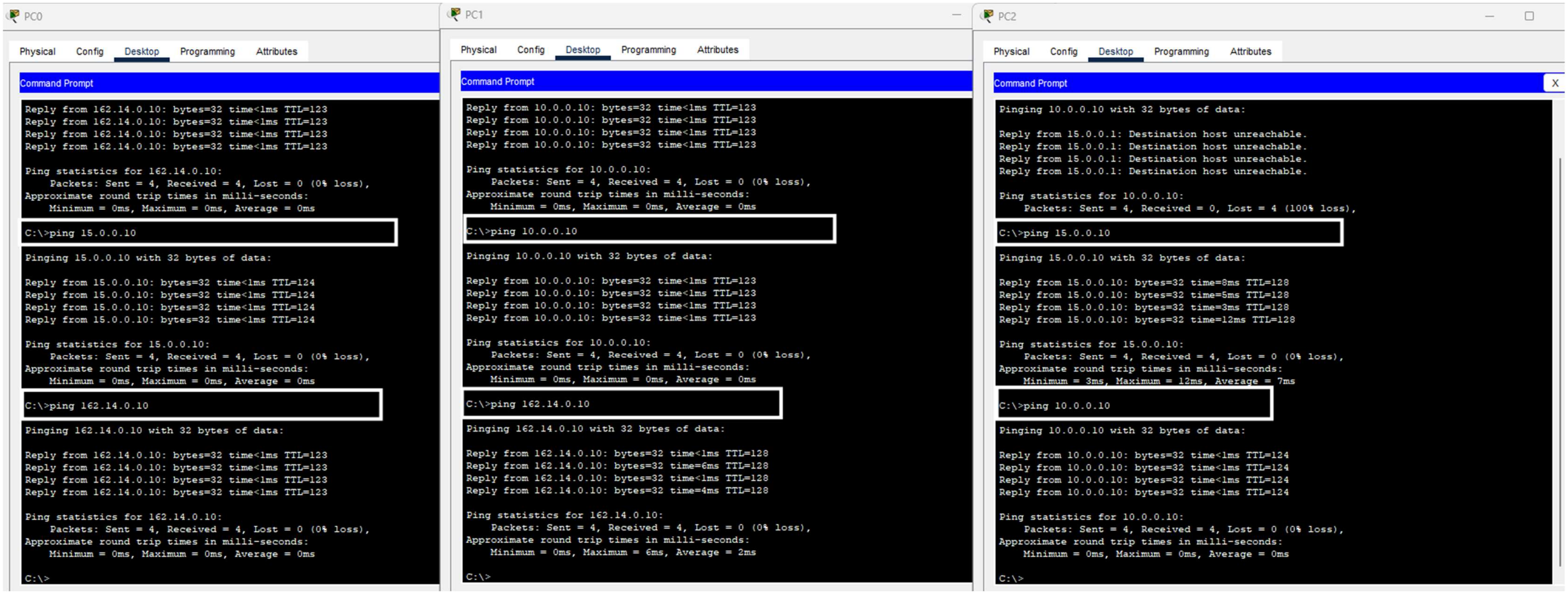
Same way add for 162.14.0.10

Static Route From 10.0.0.10 to 162.14.0.10

R1			R2			R3			R5		
Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway	Static Route	Netmask	Gateway
162.0.0.0	255.0.0.0	20.0.0.2	162.0.0.0	255.0.0.0	30.0.0.2	162.0.0.0	255.0.0.0	21.0.0.2	162.0.0.0	255.0.0.0	192.168.168.2

Static Route From 162.14.0.10 to 10.0.0.10

R5			R6		
Static Route	Netmask	Gateway	Static Route	Netmask	Gateway
10.0.0.0	255.0.0.0	21.0.0.1	10.0.0.0	255.0.0.0	192.168.168.1



Now successfully ping from each other networks.

Network Interface IP Configuration

Router	Interface IP	Prefix
R1	10.0.0.1	/8
R1	20.0.0.1	/8
R2	20.0.0.2	/8
R2	30.0.0.1	/8
R3	30.0.0.2	/8
R3	172.19.0.1	/16
R3	21.0.0.1	/8
R4	172.19.0.2	/16
R4	15.0.0.1	/8
R5	21.0.0.2	/8
R5	192.168.168.1	/24
R6	192.168.168.2	/24
R6	162.14.0.1	/16

Static Route Configuration Summary

ip route destination_network subnet_mask next_hop_ip

Traffic Flow: 10.0.0.10 → 15.0.0.10

Router	Static Route
R1	ip route 15.0.0.0 255.0.0.0 20.0.0.2
R2	ip route 15.0.0.0 255.0.0.0 30.0.0.2
R3	ip route 15.0.0.0 255.0.0.0 172.19.0.2
R4	Directly connected to the **15.0.0.0/8** network

Return Traffic: 15.0.0.10 → 10.0.0.10

Router	Static Route
R4	ip route 10.0.0.0 255.0.0.0 172.19.0.1
R3	ip route 10.0.0.0 255.0.0.0 30.0.0.1
R2	ip route 10.0.0.0 255.0.0.0 20.0.0.1
R1	Directly connected to the **10.0.0.0/8** network

Traffic Flow: 10.0.0.10 → 162.14.0.10

Router	Static Route
R1	ip route 162.0.0.0 255.0.0.0 20.0.0.2
R2	ip route 162.0.0.0 255.0.0.0 30.0.0.2
R3	ip route 162.0.0.0 255.0.0.0 21.0.0.2
R5	ip route 162.0.0.0 255.0.0.0 192.168.168.2
R6	Directly connected to the **162.0.0.0/8** network

Return Traffic: 162.14.0.10 → 10.0.0.10

Router	Static Route
R6	ip route 10.0.0.0 255.0.0.0 192.168.168.1
R5	ip route 10.0.0.0 255.0.0.0 21.0.0.1
R3	Route already exists toward the **10.0.0.0/8** network
R2	Route already exists toward the **10.0.0.0/8** network
R1	Directly connected to the **10.0.0.0/8** network

End-to-End Path

10.0.0.10 → 15.0.0.10

PC (10.0.0.10) → R1 → R2 → R3 → R4 → Host (15.0.0.10)

10.0.0.10 → 162.14.0.10

PC (10.0.0.10) → R1 → R2 → R3 → R5 → R6 → Host (162.14.0.10)